

Mingxing Weng

✉ mxweng22@sjtu.edu.com 🌐 <http://mxweng.github.io>

2911 Haijiang Avenue, Zhenhai District, Ningbo, Zhejiang Province, China 315200

Education

- ▶ **PhD in Mathematics** 2022–present
Shanghai Jiao Tong University, Shanghai, China
Advisor: Prof. Shi Jin
- ▶ **PhD in Mathematics** 2022–present
Eastern Institute of Technology, Ningbo, Zhejiang, China
Advisors: Prof. Jie Shen, Prof. Zhiping Mao
- ▶ **MSc in Mathematics (Graph Theory and Combinatorics)** 2019–2022
Nankai University, Tianjin, China
- ▶ **BSc in Mathematics** 2015–2019
Fuzhou University, Fujian, China

Research Interests

- ▶ **Computational Mathematics**
Scientific Machine Learning, Numerical PDEs, Optimization Methods, and Their Applications

Publications

- ▶ **M. Weng**, X. Liu, Z. Mao, J. Shen, *Energy-Stabilized Acceleration for Robust Optimization*, Submitted, 2026.
Summary: Development of energy-stable accelerated optimization methods for improved convergence and robustness.
- ▶ **M. Weng**, Z. Mao, J. Shen, *Deep Collocation Method: A Framework for Solving PDEs Using Neural Networks with Error Control*, SIAM Journal on Scientific Computing, 2026.
Summary: A neural-network-based framework that constructs adaptive basis functions for the efficient and accurate solution of multiscale PDEs and PDEs with low-regularity solutions.
- ▶ Q. Guo, Y. Jin, L. H. Sun, **M. Weng**, *A Semi-Bijective Algorithm for Saturated Extended 2-Regular Simple Stacks*, Advances in Applied Mathematics, 2023.
Summary: A combinatorial algorithm for counting structures related to RNA configurations.

Skills

- ▶ **Programming:** Python (PyTorch), MATLAB, Maple, LaTeX
- ▶ **Languages:** Chinese, English (Professional working proficiency), Japanese (Beginner)

Honors & Awards

- ▶ Outstanding Poster Award, CSIAM Annual Conference 2025-10
- ▶ Outstanding Undergraduate Graduate, Fuzhou University 2019-07
- ▶ National First Prize, Mathematical Contest in Modeling (Team Leader) 2017-11